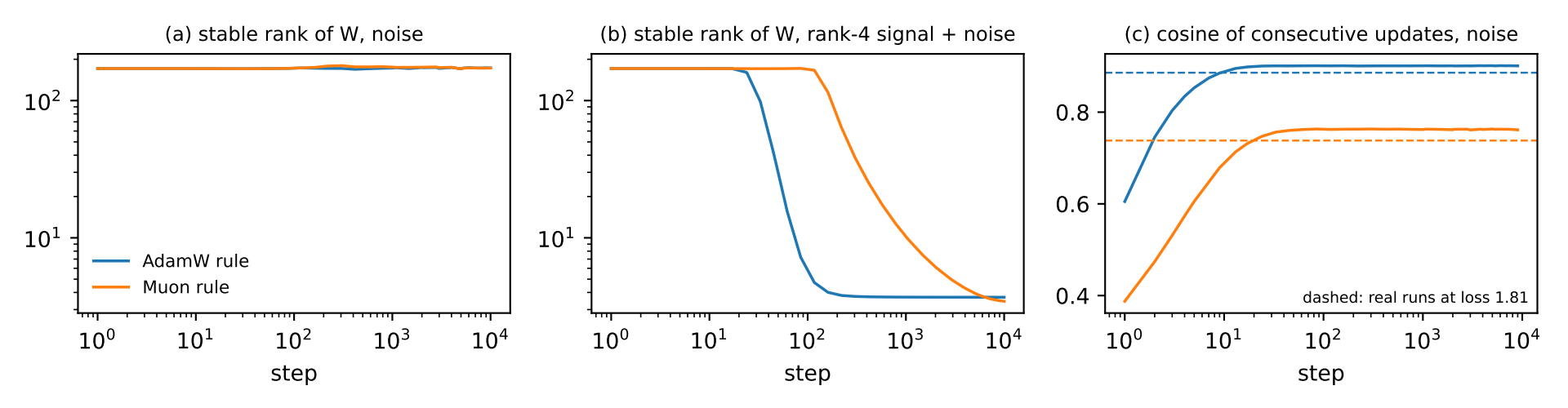
(a) stable rank of W, noise
(b) stable rank of W, rank-4 signal + noise
(c) cosine of consecutive updates, noise
10^2
10^1
0.8
0.6
0.4
AdamW rule
Muon rule
dashed: real runs at loss 1.81
10^0
10^1
10^2
10^3
10^4
step